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(12) **United States Design Patent**
Suzuki et al.

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(54) **LENS FOR LIGHT DETECTION ELEMENT**

FOREIGN PATENT DOCUMENTS

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JP 1248332 7/2006
JP 1248732 7/2006

(Continued)

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OTHER PUBLICATIONS

“Thorlabs,” release date Oct. 30, 2014, retrieved Apr. 8, 2025 from
[https://www.thorlabs.com/thorproducts.cfm?partnumber=354710-C](https://www.thorlabs.com/thorproducts.cfm?partnumber=354710-C&gQT=1)
&gQT=1 (Year: 2014).*

(Continued)

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(51) **LOC (15) Cl.** **16-06**

(52) **U.S. Cl.**
USPC **D16/134**

(58) **Field of Classification Search**
USPC D16/130, 134, 136, 137, 100, 101, 219;
D26/120, 124
CPC G02B 27/283; G02B 1/043; G02B 5/045;
G02B 3/00
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2,312,542 A 3/1943 Goodman
D206,956 S * 2/1967 Stewart D16/137
D215,481 S 10/1969 Dawson

(Continued)

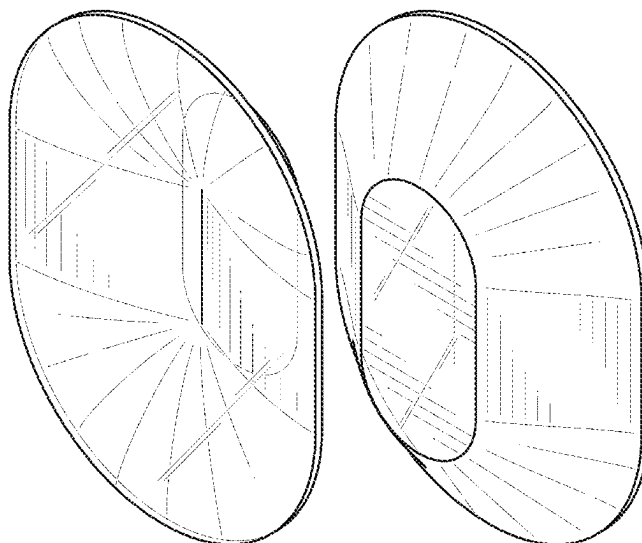
(57) **CLAIM**

The ornamental design for a lens for light detection element
as shown and described.

DESCRIPTION

FIG. 1 is a front view of a lens for light detection element
of our new design;
FIG. 2 is a rear view thereof;
FIG. 3 is a top plan view thereof;
FIG. 4 is a bottom plan view thereof;
FIG. 5 is a right side view thereof;
FIG. 6 is a left side view thereof;
FIG. 7 is a front perspective view thereof;
FIG. 8 is a rear perspective view thereof;
FIG. 9 is a sectional view taken along the line 9-9 of FIG.
1;
FIG. 10 is a sectional view taken along the line 10-10 of FIG.
1; and,
FIG. 11 is an enlarged view showing a portion of FIG. 10
defined by lines 11-11 and 11'-11'.

1 Claim, 11 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

D362,081 S 9/1995 Bilbrey
 D371,376 S 7/1996 Johnson, Jr.
 D481,050 S * 10/2003 Carr D16/101
 D489,383 S 5/2004 Overholser
 D510,945 S 10/2005 Do
 D511,787 S 11/2005 Galli
 D607,913 S 1/2010 Chen et al.
 D608,807 S 1/2010 Chern et al.
 D608,809 S 1/2010 Chen et al.
 D610,597 S * 2/2010 Wei D16/101
 D610,599 S 2/2010 Cheng et al.
 D612,880 S 3/2010 Peng et al.
 D619,755 S 7/2010 Lai et al.
 D623,674 S 9/2010 Wei et al.
 D624,105 S 9/2010 Peng et al.
 D625,045 S 10/2010 Chen et al.
 D625,462 S 10/2010 Ku et al.
 D631,184 S 1/2011 Feng et al.
 D632,322 S 2/2011 Chiang et al.
 D650,516 S 12/2011 Lai et al.
 D651,227 S * 12/2011 Lai D16/134
 D651,228 S * 12/2011 Lai D16/134
 D654,525 S * 2/2012 Lai D16/134
 D662,530 S * 6/2012 Lai D16/134
 D662,956 S * 7/2012 Lai D16/136
 D665,437 S * 8/2012 Johnson, Jr. D16/130
 D700,394 S * 2/2014 Tang D26/124
 D753,746 S * 4/2016 Yu D16/134
 D765,752 S 9/2016 Chiu
 D771,172 S 11/2016 Huang et al.
 D795,322 S * 8/2017 Kameda D16/134
 D798,357 S * 9/2017 Wei D16/134
 D830,440 S 10/2018 Mathieu

D837,857 S * 1/2019 Kameda D16/134
 D837,858 S * 1/2019 Kameda D16/134
 D846,016 S 4/2019 Iwasaki et al.
 D857,775 S 8/2019 Kang
 D859,492 S * 9/2019 Okitsu D16/134
 D863,393 S 10/2019 Iwasaki et al.
 D871,486 S * 12/2019 Cohen D16/218
 D962,320 S * 8/2022 Blommé D16/130
 D1,005,866 S 11/2023 Stephenson
 2006/0238881 A1 10/2006 Park et al.
 2009/0225440 A1 * 9/2009 Ho G02B 19/0061
 359/709
 2014/0036496 A1 * 2/2014 Cai G02B 19/0028
 362/235
 2014/0177229 A1 * 6/2014 Wang He G02B 19/0028
 362/296.01
 2014/0293582 A1 * 10/2014 Lee G02B 19/0061
 362/244
 2015/0109764 A1 * 4/2015 Nam G02F 1/133526
 362/97.3
 2024/0069314 A1 * 2/2024 Suzuki G02B 5/001

FOREIGN PATENT DOCUMENTS

JP 1250242 8/2006
 JP 2013-134985 A 7/2013

OTHER PUBLICATIONS

Notice of Allowance mailed Jun. 2, 2025 in U.S. Appl. No. 29/884,843.
 Notice of Allowance mailed Jun. 3, 2025 in U.S. Appl. No. 29/884,835.

* cited by examiner

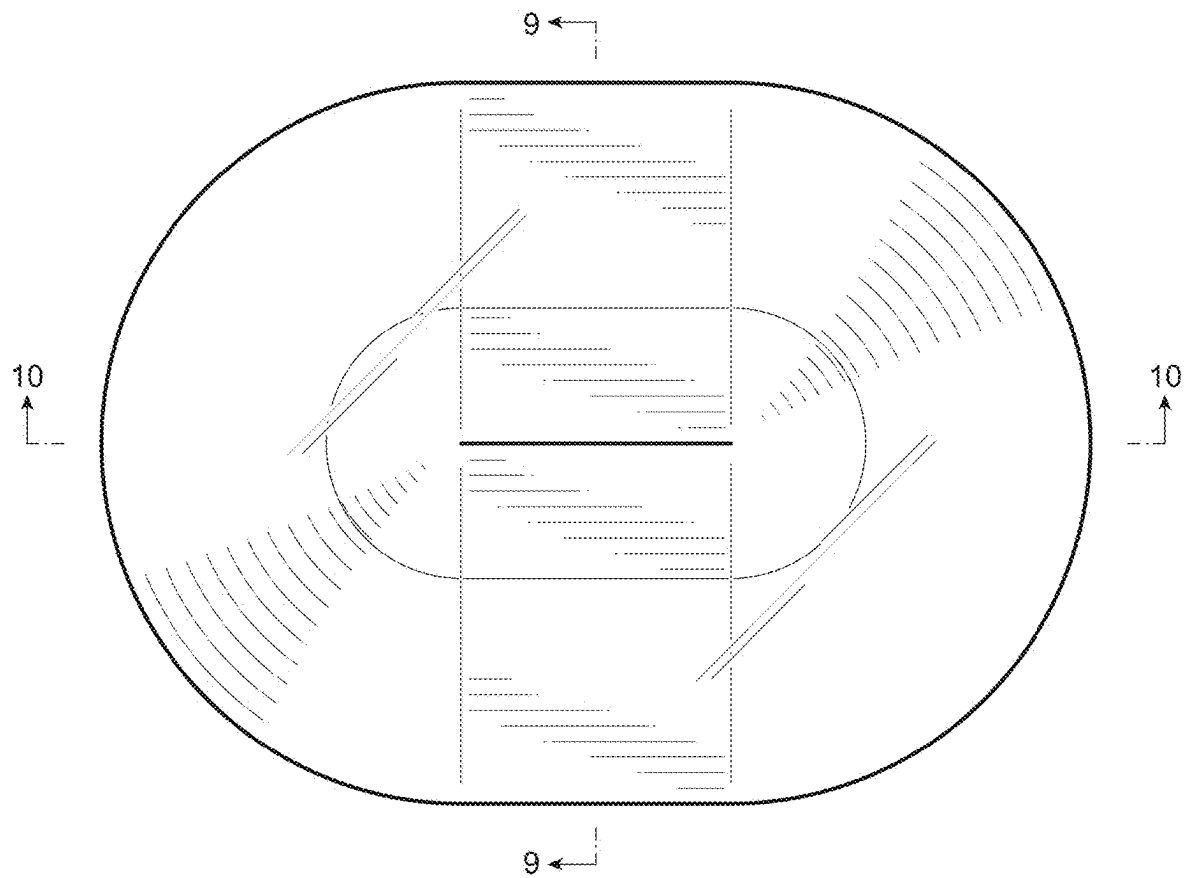


FIG. 1

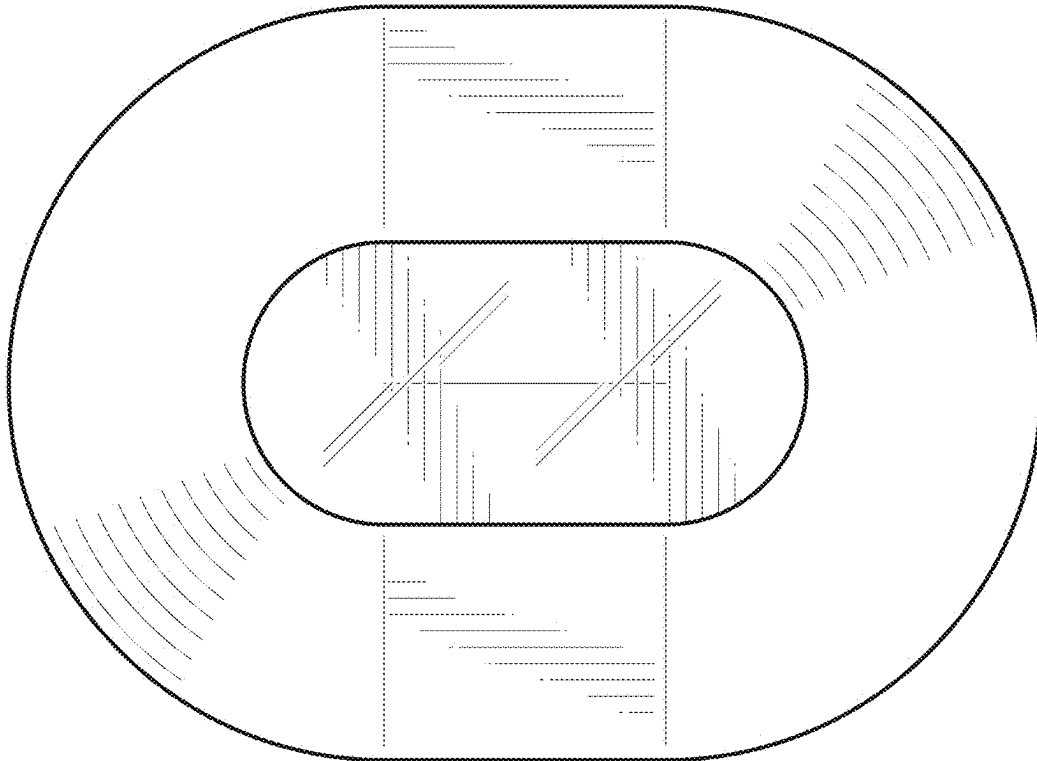


FIG. 2



FIG. 3



FIG. 4

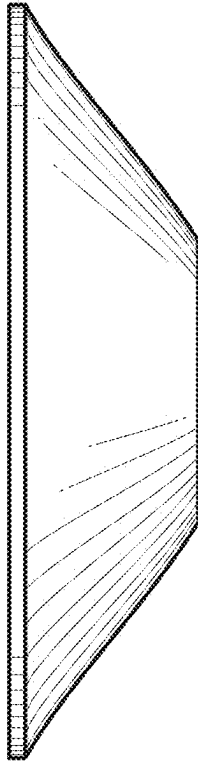


FIG. 5

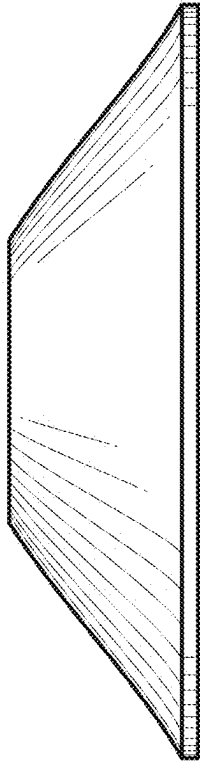


FIG. 6

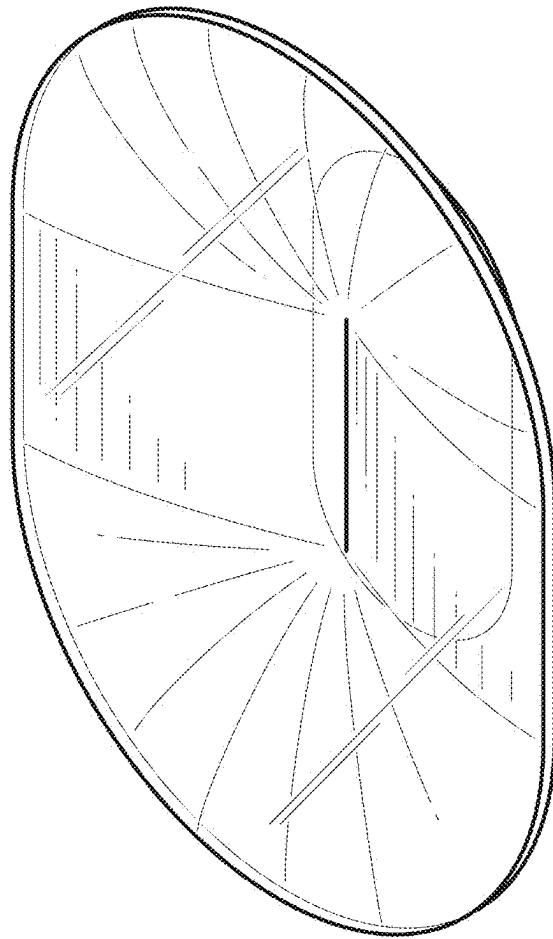


FIG. 7

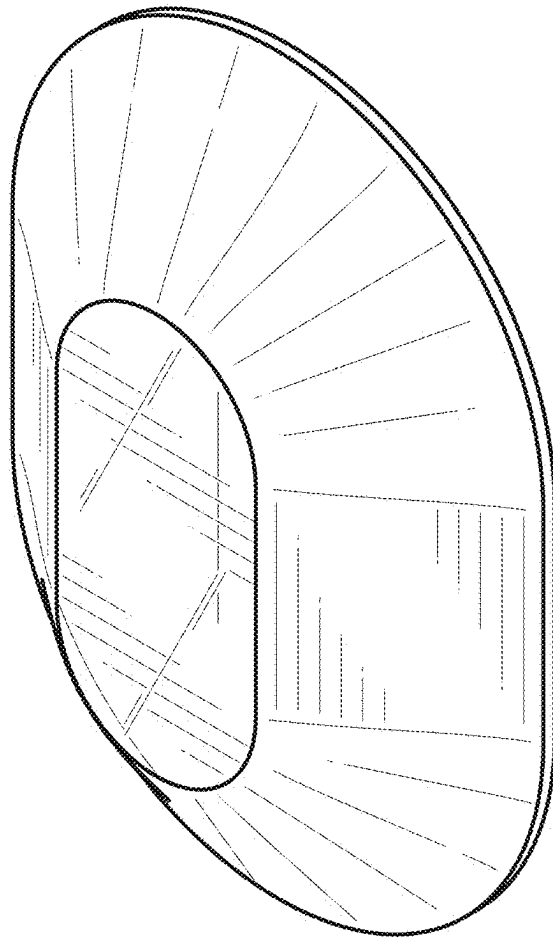


FIG. 8

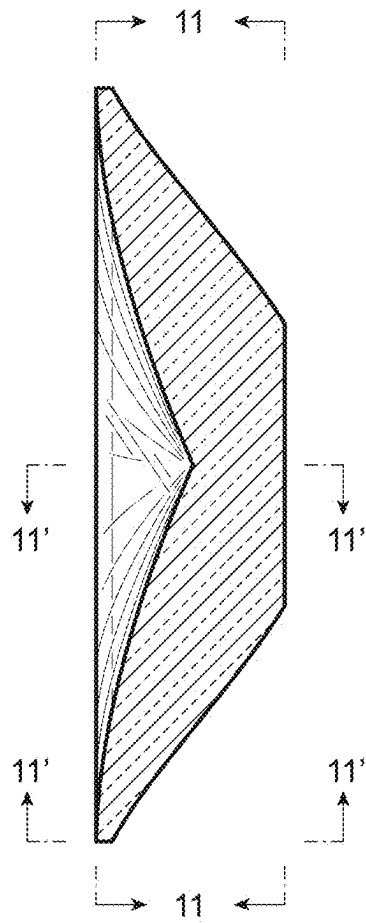


FIG. 9

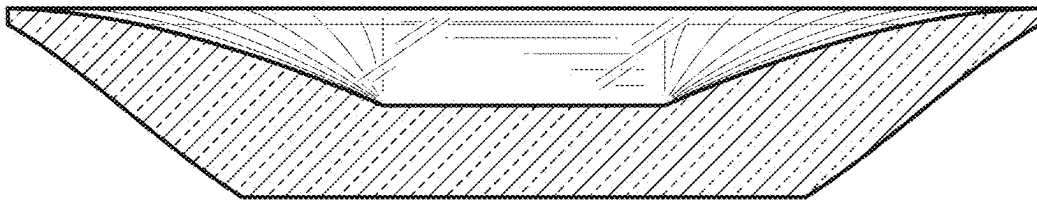


FIG. 10

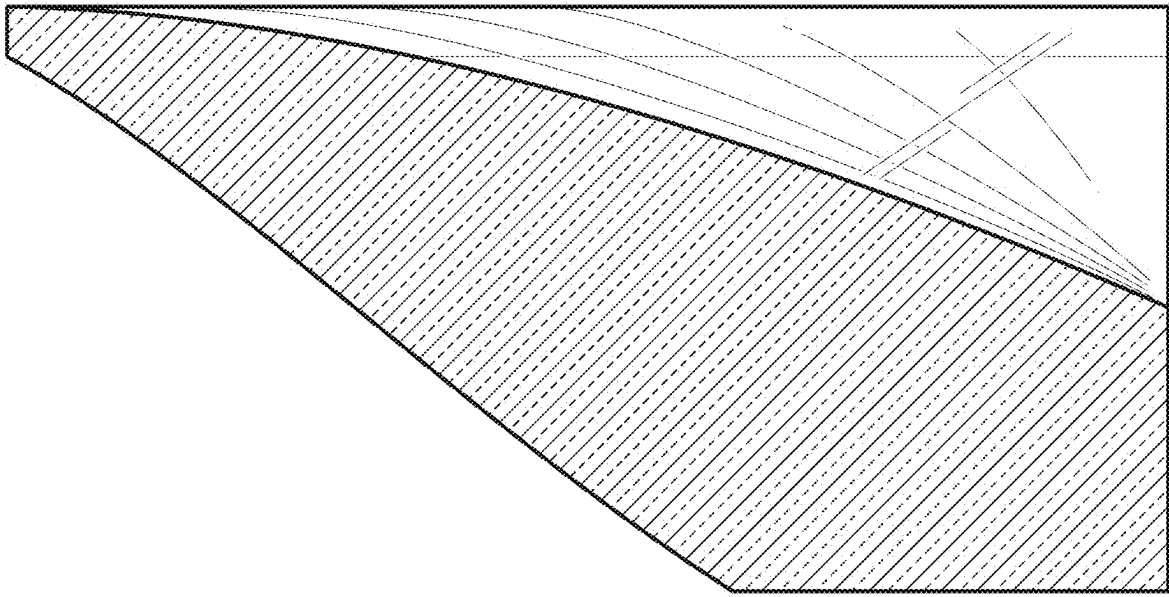


FIG. 11